

is basically stranded, insulated wire designed to make the internal connections inside our electrical device better and also improve the solderability. There are other types of wire which help in making the electromagnetic coils; which help in the close winding of the coil in the case of creating electromagnetic coils.

- 2) **Resistors:** A resistor is an important component that has a vital role in the electric current flow of the circuit. Now, this resistor plays a significant role by reducing the flow of current across the circuit. Why is this necessary? There are times the flow of current is excessive and the output we require to perform a certain task is less, then users should utilise the resistor. It works on the methodology of the Ohm's law. According to Ohm's law, voltage is equal to current multiplied with the resistor. Now, this law is followed to make use of the resistor in the circuit, and this basically helps to use the amount of current that is required. For example- if there is a circuit that is connected with a battery of 9V and you need to glow a LED of 30mA. If we pass the current through the circuit, then the LED connected to the circuit can go off or fuse. In this case, a resistor will help you to adjust the flow of current to the LED. We need to understand how much Ohm of the resistor can be used to adjust the flow. So as per the Ohm law, we will be using a 300 Ohm resistor; in this case, to manage or adjust the flow of current to light up the LED.
- 3) **Capacitors:** The capacitors are used in the circuit to store charge and then use that charge in the form of output. It basically consists of two plates - one of the plates stores the positive charge, and the other plate when connected to a supply or source stores a negative charge. Now, these two plates, when connected to a different source or supply, neutralise both the charges. This effect achieved by using the capacitor is known as capacitance.
- 4) **Inductors:** When current is flowing through a wire, a magnetic field is always created. But sometimes when there are many coils collected together during the flow of current, then, at that particular time, there is a case of the high magnetic field created. An inductor is basically temporary storage of charge that can also be used for other purposes

Security alarm circuit

If you find yourself in a situation where you need a security or an alarm system to alert you on any cases of mishandlings or theft, you can use this circuit to protect your lockers or anything which you want to keep safe. You